Skills Summary

Building Equivalent Ratio Tables

Use this reference while completing your worksheet or when studying.

A ratio table is one ratio written over and over at different sizes. Every column is the first column times some number k — the same k for both rows. Find k first, and every question on the worksheet becomes one multiplication.

1. Filling in a missing table entry

Rule: if $a : b$ is the ratio, then every column is $ak : bk$.

- Use the row where *both* entries are known to get $k = \dfrac{\text{new}}{\text{old}}$.
- Multiply the partner entry by the same k.
- Never add: $a + n : b + n$ is a *different* ratio.

Example: A table built from $3 : 4$ has a column whose first entry is 12. Find the second entry.
Step 1. The first row has both a known old and a known new value, so it is the row that reveals the scale factor.
Step 2. $k = 12 \div 3 = 4$, so the second entry is 4×4.
16

Try it: A table built from $5 : 9$ has a column starting with 35. Find its second entry.

check: **63**

2. Finding the scale factor between two columns

Rule: $k = \dfrac{\text{new entry}}{\text{old entry}}$, taken from the row where both numbers are given.

The factor does not have to be a whole number — 2.5, $\frac{3}{2}$ and 0.4 are all legal scale factors, and $k < 1$ simply scales the table down.

Example: A ratio table's first row goes from 6 to 15. What is the scale factor?
Step 1. Scale factor is a multiplier, not a difference, so divide the new entry by the old one instead of subtracting.
Step 2. $k = 15 \div 6 = \dfrac{5}{2}$, and $6 \times 2.5 = 15$ checks out.
2.5

Try it: A row of a ratio table goes from 8 to 28. Find the scale factor.

check: **3.5**

3. Adding the unit-rate column

Rule: the unit-rate column is the one whose *second* row entry is 1. Divide both entries of a known column by that column's second entry:

$$\frac{a}{b} : 1$$

The top entry of that column is the rate — "how many per one".

Example: A runner covers 12 laps in 8 minutes. Add the per-minute column.
Step 1. A unit rate is the entry paired with 1, so divide the whole column by the time entry.
Step 2. $12 \div 8 = \frac{3}{2}$ laps for $8 \div 8 = 1$ minute.
1.5

Try it: A printer prints 27 pages in 6 minutes. Find the pages-per-minute unit rate.

check: **4.5**

4. Deciding whether two tables are equivalent

Rule: write each ratio as a fraction and reduce, or compare unit rates. Two tables are equivalent exactly when the reduced fractions match.

Cross-products say the same thing: $\frac{a}{b} = \frac{c}{d}$ precisely when $ad = bc$. Use $<$, $>$, or $=$ to report the result.

Example: Is the table $6 : 9$ equivalent to the table $10 : 15$?
Step 1. Equivalent means the same ratio at a different size, so reduce both to lowest terms and compare.
Step 2. $6 : 9$ reduces to $2 : 3$, and $10 : 15$ also reduces to $2 : 3$.
$\mathbf{6/9 = 10/15}$

Try it: Compare the tables $2 : 3$ and $5 : 7$ with $<$, $>$, or $=$.

check: $\mathbf{2/3 < 5/7}$